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How Australia's commitment and activities for measles control achieved rubella elimination

A. Glynn-Robinson[1,*], J.K. Knapp[2], D.N. Durrheim[3]

[1] *Private Contractor, Narraabundah, Australia Capital Territory, Australia*
[2] *Centres for Disease Control and Prevention (CDC), Atlanta, United States*
[3] *University of Newcastle, Newcastle, Australia*

Background: The global measles and rubella strategic plan 2012–20 has a target for measles and rubella elimination in five WHO regions by the end of 2020. Only one region has eliminated rubella. The WHO Strategic Advisory Group of Experts on Immunization advised rubella control was lagging and recommended rubella vaccination be introduced into all country immunisation programmes as soon as possible. We describe the impact enhanced measles control activities, using measles-mumps-rubella (MMR) vaccines, had on rubella incidence in Australia and how these activities lead to the successful elimination of both viruses.

Methods and materials: We reviewed rubella notifications from Australia's National Notifiable Disease Surveillance System (1991–2018) and stratified by age, sex and vaccination status. Three-year cumulative incidences were calculated by birth-cohort to assess the impact of two major measles immunisation activities in Australia; the 1998 Measles Control Campaign (MCC), targeted children 5–12 years; and the 2001 Young Adult MMR immunisation program (YA-MMR), targeted adults 18–30 years. MMR immunisation coverage estimates were extracted and compared with notification data. Aggregate national serological survey data (1999, 2002, 2007 and 2012) was assigned to birth-cohorts and the mean, median and ranges of age-group estimates were calculated.

Results: The MCC attained 96% coverage, the three-year cumulative incidence of rubella declined by 90% compared with the three years preceding. This effect was also seen following the YA-MMR with a 73% decline in pre-and-post three-year periods, however vaccine coverage during this campaign was considerably lower. Serological surveys displayed a high and constant level of immune protection among females, but males were estimated to have 4–13% lower immunity. Rubella immunity significantly benefited by a direct 17% boost in serological immunity in those 6–11 years of age in the 1999 pre-and-post serological survey.

Conclusion: The use of combined MMR vaccines to address measles immunity concerns in Australia demonstrated a symbiotic benefit in not only controlling and eliminating measles transmission but rubella as well. Although the coverage of the YA-MMR was considered to be "sub-optimal", it effectively closed the rubella immunity gap in Australia, and demonstrates that utilising combination MR-containing vaccines in routine and mass vaccination programs is effective in eliminating the transmission of both viruses.

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Seroprevalence and risk factor analysis of human leptospirosis in distinct climatic regions of Pakistan

M.L. Sohail[1,*], M.S. Khan[2], M. Ijaz[3], O. Naseer[4]

[1] *Cholistan University of Veterinary and Animal Sciences, Bahawalpur, Pakistan, Bahawalpur, Punjab, Pakistan*
[2] *University of Veterinary and Animal Sciences, Lahore, Pakistan, Clinical Medicine and Surgery, Lahore, Pakistan*
[3] *University of Veterinary and Animal Sciences, Department of Clinical Medicine and Surgery, Lahore, Pakistan*
[4] *Cholistan University of Veterinary and Animal Sciences, Bahawalpur, Medicine, Bahawalpur, Pakistan*

Background: Leptospirosis is a worldwide emerging infectious disease of zoonotic importance and large epidemics and epizootics have been reported all over the globe.

Methods and materials: A cross survey study was conducted to estimate seroprevalence of human leptospirosis in climatically distinct regions of Pakistan and to identify the risk factors associated with the disease. Blood samples from 360 humans were collected through convenient sampling, 120 from each of three study areas. Serological testing was performed using ELISA kit as per manufacturer's recommendations.

Results: The results showed an overall prevalence of 40.83% (95% CI; 35.71–46.11). Statistical analysis showed significant ($p < .05$) differences in leptospiral seroprevalence in three different geographic locations, with highest in humid sub-tropical climatic region (50.83%; 95% CI; 41.55–60.07), followed by semi-arid region (44.16%; 95% CI; 35.11–53.52) and lowest in hot and dry region (27.50%; 95% CI; 19.75–36.40). After multivariate analysis age, gender, exposure to flooding water, source of water usage, disinfection schedule of surroundings and history of cut and wound were found significantly associated with the seropositivity of Leptospira.

Conclusion: The present study, first to uncover seroprevalence of human Leptospira in different climatic regions of Pakistan, alarms about effect of climate on prevalence of Leptospira in the region.

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First record of *Leishmania* spp. in birds and bats from Sudan

R. Baleela[1], E. Abdalwhab[2,*], H. Elnaim[1], M. Yousif[1], M. Shabo[1], E. Elfaki[3], M. Abdelgadir[4], N. Khalid[5]

[1] *University of Khartoum, Zoology, Khartoum, Sudan*
[2] *Ibn Sina University, Khartoum, SD, Sudan*
[3] *Wildlife Research Centre, Ministry of Animal Resources, Khartoum, Sudan*
[4] *University of Hail, Hail, Saudi Arabia*
[5] *Ibn Sina University, Zoology, Khartoum, Sudan*

Background: Sudan is one of the endemic areas of visceral leishmaniasis (VL) in the world. The disease represents a major health problem with outbreaks occurring periodically claiming the life of high number of victims. The disease is caused by many strains/species of *Leishmania* parasite which is harbored by many

different species of vertebrate reservoir hosts. The objectives of this study were to investigate the reservoir hosts of *Leishamania spp* among different vertebrates from two areas: a VL endemic area Dinder National Park (DNP) and a non-endemic area (Khartoum State); and to identify and characterize *Leishmania* spp if present.

Methods and materials: Thin blood films from 119 samples from five possible vertebrate resevoir hosts (bats, birds, Egyptian mongoose, reed buck and rodents) were microscopically examined and screened for the presence of *Leishmania* parasites. Genomic DNA was extracted from different tissue parts from the same sample. For the molecular detection of the parasite; three genes were successfully amplified: *Leishmania* minicircle kDNA, small subunit - ribosomal DNA (SSU-rDNA), and the internal transcribed spacer (ITS1) which was further characterized by restriction fragment length polymorphism (PCR-RFLP) analysis.

Results: *Leishmania* amastigotes were microscopically observed in 14.3% (3 of 21) of the bats collected from DNP. *Leishmania* specific genomic target DNA was amplified in 17.6% of the sample (21 of 119) belonging to five vertebrate species: the Egyptian mongoose *Herpestes ichneumon,* the Spiny mice *Acomys albegina*, the Nile rat *Arvicanthis nilotucs*, the Reed Buck *Redunca arundinum,* birds including (the African jacana *Actophiloms africanus*, Pied kingfisher *Cerylr rudis,* greater painted snipe *Rostratula benghalensis)* and two bats: *Glauconycteris variegata* and *Scotophilus nigrita*. The ITS1-RFLP analysis successfully identified *Leishmania donovani* in (2 out of 5) positive samples of bats.

All positive samples were collected from DNP.

Conclusion: The detection of *Leishmania* spp. in bats, birds and reedbuck is the first of its kind in Sudan. The finding of *L. donovani* in bats from Sudan has serious implications on control programs. More work is needed for further identification of the *Leishmania* species detected in birds and the reedbuck, in addition to the evaluation of the capacity of these animals to transmit the disease.

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Evaluation of pneumonia in children under five surveillance system, Savelugu-Nanton Municipality, Northern Region, Ghana, 2019

A. Baffour Appiah[1,*], S. Dapaa[2], C. Kubio[3], B.B. Kaburi[4], D.K. Ameme[4], E. Kenu[1]

[1] *Ghana Field Epidemiology and Laboratory Training Programme, Department of Epidemiology and Disease Control, School of Public Health, University of Ghana, Acccra, Ghana*
[2] *Ghana Field Epidemiology and Laboratory Training Programme, School of Public Health, University of Ghana, Legon, Accra, Accra, Ghana*
[3] *Karaga District Health Directorate, Ghana Health Service, Karaga, Ghana*
[4] *Ghana Field Epidemiology and Laboratory Training, Accra, Ghana*

Background: An estimated 920,136 children died worldwide from pneumonia in 2015. In Ghana, about 4000 children die each year from pneumonia, which makes it the 5th leading cause of death. The Northern Region tops under five mortality rate in Ghana, and pneumonia is the third leading causes of deaths in Northern Region. The pneumonia under-five surveillance system has not been evaluated since its inception in 2012 in the Northern region of Ghana. We evaluated the pneumonia under-five surveillance system in Savelugu-Nanton Municipality, to examine whether the system was meeting its objectives.

Methods and materials: We conducted this evaluation in Savelugu-Nanton Municipality from January to February, 2019. Systems' performance, usefulness and attributes were assessed using CDC updated guidelines (2001). Surveillance data in the district health information management system (DHIMS 2) at the municipality from 2015–2018 were extracted and analyzed descriptively using frequencies and proportions, and presented in tables and figures. We interviewed stakeholders using semi-structured questionnaire. Records were also reviewed at some health facilities, sub-municipal and at municipal levels.

Results: A total of 3351 cases were detected and reported over four year period but cases were not classified (as severe and non-severe), and alert epidemic threshold does not exist. Males (54.2%) and age bracket 1–4 years (56.1%) were mostly affected. Pneumonia under-five peaked in 2016 (3073.3 per 100,000) and average rate of 2575.1 per 100,000. The system was easy to operate since it was integrated with other diseases and does not require filling case based form. The system was operating at all levels but data disparity existed between DHIMS 2 and some reporting units. Predicted value positive was not estimated since laboratory and/or X-ray confirmation were lacking.

Conclusion: The pneumonia under-five surveillance system is useful and partly achieving its objectives. The system was simple, sensitive, stable, acceptable, flexible and representative. Data quality was good. We recommend that laboratory component should be strengthened and to provide additional columns on the Intergrated Disease Surveillance System and Response (IDSR) reporting forms and in DHIMS 2 for reporting severe and non-severe cases.

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Identification of Lumpy skin disease (LSD): First-time in Bangladesh during the investigation of unknown skin disease of cattle

F. Talukdar[1], A. Fakir[2], Z. Hasan[3], T.M.G. Osmani[2], G.A. Choudhury[4], S. Budhay[4], M. Sadekuzzaman[4], E. Brum[5], H.R. Bhowmic[2], B. Noorjahan[2,*]

[1] *IEDCR, DLS, Dhaka, Bangladesh*
[2] *DLS, DLS, Dhaka, Bangladesh*
[3] *Food and Agriculture Organization of the United Nations, ECTAD, Dhaka, Bangladesh*
[4] *CDIL, DLS, Dhaka, Bangladesh*
[5] *Food and Agriculture Organization of the United Nations, ECTAD Bangladesh, Dhaka, Bangladesh*

Background: Cattles are often infected with different skin diseases but mostly without no serious consequences. In July 2019, District Livestock Officer of Chattogram reported an unknown skin disease in cattle which was spreading rapidly with casualty. Comparative studies and effective control of such skin disease in cattle requires accurate laboratory techniques to confirm provisional clinical diagnosis. The aim of this study is to determine the extent and causes of the outbreak as well as controlling the skin disease by effective investigation of skin lesions.

Methods and materials: To facilitate the study of the reported skin disease pathogenesis, we selected the livestock population from different areas of Chattogram district (Patia, Karnafully and Anowara upazilla). We interviewed the farm owners and used different physical examination techniques to investigate the outbreak. Cattle developing skin nodules especially in the head and neck, per-